

Telecom Customer Support Knowledge Base

Support procedures for broadband mobile billing accounts and installation

This manual provides practical guidance for telecom support agents. Equipment-specific instructions, contractual entitlements, identity-verification requirements and financial adjustments must follow the applicable approved procedure. It does not establish new customer charges, credit limits, restoration guarantees or regulatory requirements.

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Chapter 1 Fiber Broadband Gateways and Wi-Fi Troubleshooting

1.1 Understand the equipment

The Optical Network Terminal, or ONT, connects the premises to the fibre access network. The gateway or router connects that service to local wired and wireless devices. Some installations combine functions in one unit.

A healthy optical connection does not by itself establish that authentication, addressing, routing or Wi-Fi is working.

1.2 Read ONT lights correctly

Identify the exact manufacturer and model before interpreting colours. Some devices have separate Power, PON and LOS indicators; others use a combined Optical light.

The following patterns illustrate common Openreach equipment, not a universal chart for every ONT. [1]

Power light showing Green

On the referenced ONT models, equipment has power. Confirm the exact model guide before taking action.

Power light showing Off

On the referenced ONT models, check the accessible approved power connection and supply. Confirm the exact model guide before taking action.

PON light showing Solid green

On the referenced ONT models, optical registration is established. Confirm the exact model guide before taking action.

PON light showing Flashing with LOS off

On the referenced ONT models, registration or verification may be in progress. Confirm the exact model guide before taking action.

PON light showing Off

On the referenced ONT models, optical connection or registration needs investigation. Confirm the exact model guide before taking action.

LOS light showing Red or flashing red

On the referenced ONT models, this pattern commonly indicates an optical alarm; confirm the model's meaning. Confirm the exact model guide before taking action.

LAN or Port light showing On

On the referenced ONT models, ethernet link is detected. Confirm the exact model guide before taking action.

LAN or Port light showing Flashing

On the referenced ONT models, this pattern usually indicates Ethernet activity. Confirm the exact model guide before taking action.

LAN or Port light showing Off

On the referenced ONT models, check the supported Ethernet connection. Confirm the exact model guide before taking action.

A green or flashing-green LOS light must not be assigned a universal meaning. Confirm the printed label and model guide; the customer may be describing a PON, Optical or Port indicator.

Record the light pattern before any restart. If registration does not complete within the documented startup period, escalate according to the equipment procedure.

1.3 Safe power and physical checks

1. Ask whether the ONT and gateway are powered.
2. Check accessible connections to the supplied, approved adapters.
3. Ask about recent furniture movement, drilling, building work or power interruptions.
4. Request a safe visual observation of the cable route.
5. Look for reported crushing, sharp bends, pulling, visible damage or water exposure.
6. Escalate suspected optical damage to qualified field support.

Do not direct a customer to disconnect fibre ends or attempt optical cleaning.

If there is smoke, a burning smell, liquid ingress or visible electrical damage, keep the equipment out of use. Any disconnection must be safe; otherwise, the customer should avoid contact and obtain qualified assistance.

1.4 WAN port and Ethernet checks

The gateway's WAN port connects it to the upstream service. Its LAN ports connect local devices.

1. Confirm which ONT Ethernet port is provisioned.
2. Confirm that its lead reaches the gateway's designated WAN port.
3. Check the link indicators at both ends.
4. With consent, reseal only accessible Ethernet connectors.
5. Test a known-good compatible Ethernet lead where available.
6. Check supported negotiated link speed and errors.

If optical registration is healthy but Ethernet link is absent, investigate the lead, supported port and equipment before assuming a fibre-line fault.

If Ethernet is healthy but internet remains unavailable, check the authorised service state, authentication, WAN addressing and routing.

1.5 Diagnosing evening drops

Evening timing can suggest increased usage, but it does not prove network congestion.

1. Record the approximate times and duration of interruptions.
2. Establish whether all devices fail together.
3. Compare wired and wireless behaviour during the affected period.
4. Check whether the gateway restarts or only loses its internet session.
5. Review accessible service incidents and authorised diagnostics.
6. Check local uploads, backups, access schedules and mesh performance.

7. Compare measurements during affected and unaffected periods.

Wired and wireless devices fail together

Access service, gateway or WAN session.

Only wireless devices fail

Radio environment, coverage or mesh.

Gateway uptime resets

Power, thermal condition, software or hardware.

Optical alarms coincide with drops

Fibre access or optical equipment.

Heavy uploads coincide with poor responsiveness

Local bandwidth saturation.

Exact scheduled interruption

Access policy or configured equipment schedule.

Escalate with timestamps, affected devices, light observations and test results. Do not prescribe a firmware version unless it is verified for that model and fault.

1.6 Wi-Fi dead zones

Walls, metal objects, interference and equipment placement can reduce wireless performance. Wireless bands also differ in their reach and sensitivity to obstacles. [2]

1. Compare service near the gateway with the affected location.
2. Test another compatible device.
3. Check whether the issue is weak coverage, interference or a device-specific fault.
4. Place the gateway in an open, suitable location.
5. Avoid enclosed cupboards, metal barriers and unnecessary obstruction.
6. Review supported band and channel settings.
7. Retest where the customer actually uses the service.

Do not promise full coverage in every room without an assessment.

1.7 Mesh puck placement

A satellite needs a reliable connection to the main mesh unit. Placing it deep inside the existing dead zone may leave it with poor backhaul.

1. Confirm that the main unit works normally.
2. Check the satellite's connection quality using its supported application.
3. Move it toward an area with a stronger connection to the source unit.
4. Keep it in an open position away from major obstructions.
5. Test service beyond the satellite.
6. Consider supported wired backhaul where suitable cabling is available.

Follow the specific mesh system's placement and wiring guidance. [3]

1.8 Router overheating

Restricted ventilation and direct sunlight can contribute to overheating and degraded operation. [4]

Ask whether the gateway is:

- Covered or enclosed.
- Stacked against other hot equipment.
- Exposed to direct sunlight.
- Restarting or becoming unresponsive.

For equipment without a safety warning, follow approved guidance to improve ventilation and observe stability. Warmth alone does not establish hardware failure.

For a burning smell, smoke, deformation or electrical damage, stop ordinary troubleshooting and arrange the designated safety response.

1.9 Wired speed testing

1. Confirm the subscribed service and any applicable documented restrictions.
2. Use a compatible computer and Ethernet connection to a gateway LAN port.
3. Confirm that the device, adapter and link support the expected test rate.
4. Turn off that computer's Wi-Fi so the test uses Ethernet.
5. With consent, pause competing transfers for the measurement.
6. Use the approved test endpoint.
7. Record download, upload and relevant latency results.
8. Repeat sufficiently to establish whether the result is consistent.

Do not disconnect the gateway and attach a computer directly to the ONT unless the service-specific procedure authorises that test.

Link speed, wireless advertising rates and measured internet throughput are different quantities. Applications may also display MB/s, while service plans use Mbps. [5]

1.10 Compare wired and wireless results

Wired normal and Wi-Fi poor

Investigate wireless placement, interference, mesh and client capabilities.

Both poor across multiple devices

Investigate service profile, local load, gateway and access performance.

One device poor

Investigate its adapter, software or application settings.

One remote destination poor

Investigate that destination or path without declaring the whole line faulty.

Use the applicable service terms when assessing performance. Do not promise every application will achieve the advertised access rate.

1.11 Restart versus factory reset

A restart and a factory reset are different actions.

- Restart temporarily interrupts operation.
- Factory reset may erase customised or required settings.

A factory reset is not the default response to disconnections. Use it only when the approved procedure requires it, consent is obtained, and the supported restoration method is confirmed.

Chapter 2 Mobile Services SIM and Porting Procedures

2.1 Begin with service scope

Establish whether the problem affects:

- Incoming or outgoing calls.
- Messages.
- Mobile data.
- One location or several.
- One handset or several subscribers.

Check account eligibility, outages, coverage and supported device configuration before replacing a SIM.

Manufacturer-supported checks can include refreshing network registration, restarting the device, checking that the intended line is enabled and installing supported carrier settings. [6]

2.2 Dropped-call investigation

1. Record safe location and approximate time information.
2. Determine whether calls drop while stationary or moving.
3. Compare indoor and outdoor experience.
4. Check known incidents and handset compatibility.
5. Review whether voice and data fail together.
6. Where authorised, compare another supported handset or SIM arrangement.
7. Escalate reproducible network faults with non-sensitive call examples.

Never ask a customer to operate a handset while driving or repeatedly dial emergency services as a test.

2.3 VoLTE and supported voice settings

VoLTE provides voice over LTE on supported networks and devices. Availability depends on the handset, software, account and network configuration.

1. Verify handset and carrier compatibility.
2. Check the intended line and voice entitlement.
3. Follow the exact manufacturer instructions for available voice settings.
4. Apply only approved carrier or software updates.
5. If data works but voice fails, investigate voice provisioning and registration.
6. Verify ordinary incoming and outgoing calls.

Do not assume every handset exposes a VoLTE switch. Avoid hidden engineering menus and unsupported forced network modes.

2.4 Indoor signal dead zones

1. Compare indoor and nearby outdoor service.
2. Check whether other supported devices show the same pattern.
3. Review network coverage and incidents.
4. Assess supported Wi-Fi Calling where eligible.
5. If the issue persists, request the appropriate coverage assessment.

Explain that enabling Wi-Fi Calling, where supported, does not repair outdoor radio coverage. Do not promise a free booster or recommend an unapproved repeater.

2.5 eSIM activation errors

Record the exact error: expired code, previously used code, unsupported device, missing line or incomplete provisioning.

1. Verify the requester through the secure process.
2. Confirm device compatibility and any applicable carrier lock.
3. Check the accepted order and intended device.
4. Establish whether the profile is absent, installed but disabled, or installed but unable to register.
5. Follow supported manufacturer checks.
6. Review activation status before retrying or reissuing.
7. Verify the intended line's calls, messages and data.

Apple's guidance includes checking the displayed line, refreshing it where appropriate, restarting and reviewing carrier settings. These instructions are device-specific. [7]

Do not delete an existing working profile merely to try again. Reissue or transfer must follow the authorised process.

2.6 Physical SIM faults

For No SIM or suspected card damage:

1. Follow the handset's approved tray procedure.
2. Do not force the tray, trim a card or use an unsuitable adapter.
3. Where permitted, compare a compatible authorised handset.
4. Distinguish card failure from tray or handset failure.
5. Replace a confirmed faulty SIM only after the required verification.
6. Confirm the old and replacement profile states.

For PIN or PUK prompts, stop code guessing and use the secure official recovery process.

2.7 Number-porting delays

1. Verify the authorised transfer request.
2. Confirm the submitted subscriber details.
3. Check the actual order state and any rejection reason.
4. Compare progress with the provider's documented transfer arrangements.
5. Investigate detail mismatches, conflicting orders or incomplete routing.
6. Correct only verified errors through the porting team.
7. Provide confirmed progress, not an invented completion time.

Do not advise cancelling the donor service prematurely; follow the approved transfer procedure.

After completion, verify incoming calls, outgoing identification, messages and data. Partial routing problems need investigation even if some calls already work.

2.8 Mid-cycle upgrades and proration

Before accepting a change, explain:

- The eligible effective date.
- New recurring charges.
- Allowance and benefit changes.
- Any confirmed contractual consequences.
- Applicable partial-period treatment.

Providers may apply partial-month charges or credits when plans or add-ons change, but another provider's practice does not establish the telecom provider policy. [8]

Agent procedure:

1. Review the current plan and documented change terms.
2. Confirm whether immediate or next-cycle fulfillment is eligible.
3. Preview the applicable billing treatment.
4. Explain charges and credits separately.
5. Obtain explicit acceptance.
6. Process the eligible order.
7. Confirm the effective date and next-bill explanation.

Illustrative calculation only: under a policy using actual days in a 30-day cycle, a change after 10 days might allocate 10/30 of the old monthly rate and 20/30 of the new rate. Do not use this calculation when the actual policy uses different dates, rounding, allowance treatment or advance-billing rules.

Chapter 3 Billing Audits Portal Access and Security

3.1 Auditing an itemised bill

1. Verify the account holder or authorised representative.
2. Identify the disputed amount and period.
3. Compare current and previous statements.
4. Review accepted orders and effective dates.
5. Separate recurring charges, usage, equipment, one-time fees, taxes and adjustments.
6. Check payment allocation and carried-forward balances.
7. Compare relevant usage and consent records.
8. Determine whether the charge is valid, erroneous or still under investigation.
9. Explain the evidence and next action.

A higher bill is not automatically an incorrect bill. Conversely, an unfamiliar charge should not be dismissed without checking its basis.

3.2 Roaming fee disputes

A customer describing roaming as accidental does not automatically establish refund entitlement.

1. Review the applicable plan and destination coverage.
2. Check the recorded network, timestamps and chargeable usage.
3. Investigate possible rating errors, unsupported charges or disputed consent.
4. Explain verified findings in plain language.
5. Reverse a confirmed error only through authorised billing controls.
6. Submit a valid-charge goodwill request separately, where policy permits.
7. Confirm the approved adjustment and its posting state.

A border-network registration or background data session may require explanation and review. Do not assume the outcome without evidence.

3.3 Courtesy credits

A courtesy credit is a discretionary service-recovery decision, separate from correcting an erroneous charge or applying a documented entitlement.

Before issuing one:

- Confirm eligibility.
- Check the agent's approval authority.
- Review previous relevant adjustments.
- State the proposed amount and reason accurately.
- Obtain required approval.
- Record whether it is pending or posted.

Never invent a standard credit amount, promise approval in advance or state that a credit has reached the bill before verifying it.

3.4 Payment not reflected

1. Verify the account through the approved process.
2. Obtain the permitted payment reference through a secure channel.
3. Distinguish pending, settled, failed and reversed transactions.
4. Compare settlement records with the subscriber ledger.
5. Reconcile confirmed misallocations through billing.
6. Check restoration eligibility if service is restricted.
7. Verify the resulting balance and service state.

Avoid instructing customers to make duplicate payments simply because reconciliation is incomplete.

3.5 Portal login lockouts

First distinguish:

- Forgotten credentials.
- Account lockout.
- MFA failure.
- Browser-specific fault.
- Portal outage.
- Insufficient user permissions.

Then:

1. Check portal incident status.
2. Confirm the exact error without collecting secrets.
3. Use the approved recovery or unlock route.
4. Investigate browser-specific behaviour with supported comparisons.
5. Retain security restrictions where a risk review requires them.
6. Verify normal access after the authorised action.

Do not change a registered contact address solely because the requester cannot access the old one.

3.6 MFA recovery

Troubleshooting and resetting MFA are different processes.

For an authenticator problem, check supported device time settings and ordinary application operation before assuming the authenticator must be replaced.

When access to an authenticator is lost:

1. Use the documented recovery route.
2. Verify ownership at the required assurance level.
3. Use an existing approved recovery method or repeat identity verification as required.
4. Bind a replacement authenticator only after authorisation.
5. Revoke obsolete authenticators or sessions as appropriate.
6. Notify the customer through approved channels.
7. Verify new sign-in and account activity.

NIST distinguishes account recovery from ordinary authentication and describes recovery methods that can include recovery mechanisms or repeated identity proofing. It also requires notifications for relevant recovery events. [9, 10]

3.7 Mandatory KYC

The acceptable evidence and checks must come from the approved requirements for the service and jurisdiction.

Procedure:

1. Identify the required verification workflow.
2. Explain its purpose and secure submission route.
3. Confirm authorised applicant or representative status.
4. Check required evidence for readability, validity and consistency.
5. Complete the prescribed authenticity and identity checks.
6. Route failures to the approved resubmission or specialist-review process.
7. Activate or restore only the functions authorised by the verified result.

Do not bypass KYC because a customer is upset or because optical character recognition failed. A readable document and a successful identity check are different findings.

Do not request identity documents in ordinary chat or retain them outside the designated process.

3.8 Suspected account compromise

1. Escalate through the security response process.
2. Verify ownership using the approved secure route.
3. Apply authorised containment.
4. Review suspicious access, contact changes and relevant service orders.
5. Restore only verified account details.
6. Revoke compromised access where required.
7. Notify the customer through approved channels.
8. Confirm the follow-up owner and next update.

Limit disclosure of unrelated customers' information if an account-linking fault is discovered.

Chapter 4 Home Installation and Technician Dispatch

4.1 Pre-installation readiness

Before confirming readiness, establish:

- Correct service address and accepted order.
- Eligible service availability.
- An authorised person able to approve access and work.
- Required landlord or building permissions.
- Accessible approved power.
- Proposed ONT and gateway locations.
- Safe internal and external access.
- Any dependency involving alarms, telecare or existing voice services.

Openreach's installation guidance illustrates the importance of permissions, access, power and an agreed cable route. Provider-specific attendance requirements must be checked separately. [11]

4.2 Drop-cable entry

The technician must assess the route. An agent should not prescribe drilling, excavation or physical construction instructions to an unqualified customer.

The authorised route should account for:

1. Required access and permissions.
2. Existing utilities and structural restrictions.
3. Manufacturer-specified fibre handling and bend limits.
4. Protection against crushing, pulling and exposure.
5. Applicable sealing and fire-safety requirements.
6. The agreed internal termination location.

Where additional work is needed, explain the verified prerequisite and planning owner. Do not keep booking standard visits for a site that is not ready.

4.3 ONT wall mounting

Installation must follow the exact equipment instructions and applicable site requirements.

Qualified technicians should:

- Use an approved mounting arrangement and suitable surface.
- Maintain specified orientation and ventilation clearances.
- Avoid locations exposed to water, damaging heat or impact.
- Provide supported accessible power.
- Protect the fibre route without sharp bends or strain.
- Leave access for authorised maintenance.
- Agree a practical router connection and placement.

Do not invent a universal mounting height, screw type, fibre bend radius or clearance. Those values depend on the model and installation specification.

Customers should not detach, relocate or remount an installed ONT themselves.

4.4 Installation completion

Before recording completion:

1. Confirm the authorised physical work is finished.
2. Verify equipment registration and service provisioning.
3. Check the optical and Ethernet states.
4. Test the supported customer connection.
5. Explain power, routine restart and support arrangements.
6. Identify any remaining defect or incomplete item.
7. Record the accepted handover.

Normal indicator lights alone do not demonstrate that the complete service works.

4.5 Missed technician appointments

1. Confirm the accepted appointment window.
2. Review dispatch and visit records.
3. Establish what happened without prematurely blaming the customer or technician.
4. Check verified availability.
5. Offer an eligible confirmed replacement slot.
6. Assess any documented compensation entitlement separately.
7. Confirm the new appointment and access requirements.

Suggested wording:

I can see the agreed visit did not complete. I'll check the dispatch record and available replacement appointments. I'll also review whether an adjustment applies under the documented terms.

Do not guarantee next-day service, priority attendance or a credit unless confirmed.

4.6 Truck-roll fee disputes

A truck-roll fee is a charge associated with dispatching a technician. Its validity depends on the accepted terms and recorded circumstances.

Review:

- Whether the fee was disclosed and applicable.
- The accepted appointment and visit outcome.
- Any agreed cancellation or access conditions.
- The technician's findings.
- Whether the charge was duplicated or allocated incorrectly.
- Any documented waiver or entitlement.

Procedure:

1. Verify the customer and disputed fee.
2. Obtain the visit and booking records.
3. Compare the reason charged with the applicable terms.
4. Investigate conflicting evidence.
5. Correct a verified error through billing.
6. Refer a valid-charge discretionary waiver for approval where permitted.

7. Explain the outcome and revised balance.

A customer-side fault does not automatically establish a fee; a provider-side fault does not establish a universal waiver without reviewing the terms.

4.7 Unsafe or incomplete visits

If a technician records unsafe access, missing permission, unavailable capacity or required preparatory work:

1. Respect the recorded stop condition.
2. Clarify the verified prerequisite.
3. Assign the appropriate planning, dispatch or site-contact owner.
4. Confirm readiness before rebooking.
5. Share only verified milestones with the customer.

Do not pressure a technician to override a safety assessment.